

HUM 4201: PRINCIPLES OF ECONOMICS

Md. Farid Dewan, Assistant Professor

Department of Economics, NSTU

Chapter-03:

Unemployment

MOTIVATION

Key Theme

- Unemployment is one of the most important macroeconomic problems.
- Losing a job can be among the most distressing economic events in a person's life.
- Most people depend on labor income to maintain their standard of living.
- Employment also provides:
 - ✓ Financial security
 - ✓ Personal accomplishment
 - ✓ Social identity
 - ✓ Psychological well-being

Why Study Unemployment?

- Affects individuals, households, firms, and the entire economy.
- Politicians often focus on job creation because employment strongly influences voters' welfare.

MOTIVATION

Why Unemployment Matters

Economic Consequences

When people who want to work cannot find jobs:

- National production decreases.
- Valuable labor resources remain idle.
- GDP falls below potential.
- Household incomes decline.
- Poverty and inequality may increase.

Personal Consequences

- Lower current living standards
- Anxiety about future income
- Reduced self-esteem
- Psychological stress

Unemployment is both an economic problem and a social problem.

IDENTIFYING UNEMPLOYMENT

Employed:

- This category includes those who worked as paid employees, worked in their own business, or worked as unpaid workers in a family member's business.
- Both full-time and part-time workers are counted.
- This category also includes those who were not working but who had jobs from which they were temporarily absent because of, for example, vacation, illness, or bad weather.

Unemployed

- This category includes those who were not employed, were available for work, and had tried to find employment during the previous four weeks.
- It also includes those waiting to be recalled to a job from which they had been laid off.

Not in the labor force

- ✓ This category includes those who fit neither of the first two categories, such as full-time students, homemakers, and retirees.

IDENTIFYING UNEMPLOYMENT

Labor Force:

- The total number of workers, including both the employed and the unemployed

$$\text{Labor Force} = \text{Employed} + \text{Unemployed}$$

Unemployment Rate

- The percentage of the labor force that is unemployed

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed}}{\text{Labor Force}} \times 100$$

Labor-force Participation Rate

- The percentage of the adult population that is in the labor force

$$\text{Labor Force Participation Rate} = \frac{\text{Labor Force}}{\text{Adult Population}} \times 100$$

IDENTIFYING UNEMPLOYMENT

Example

January 2016 Data of USA

Employed = 150.5 million

Unemployed = 7.8 million

Adult Population = 252.4 million

Calculating Labor Force

Labor Force = $150.5 + 7.8 = 158.3$ million

Interpretation

- 158.3 million people were participating in the labor market.

Calculating Unemployment Rate

$$\text{Unemployment Rate} = \frac{7.8}{158.3} \times 100 = 4.9\%$$

Meaning

- Approximately 5 out of every 100 labor-force participants were unemployed.

Calculating Labor-Force Participation Rate

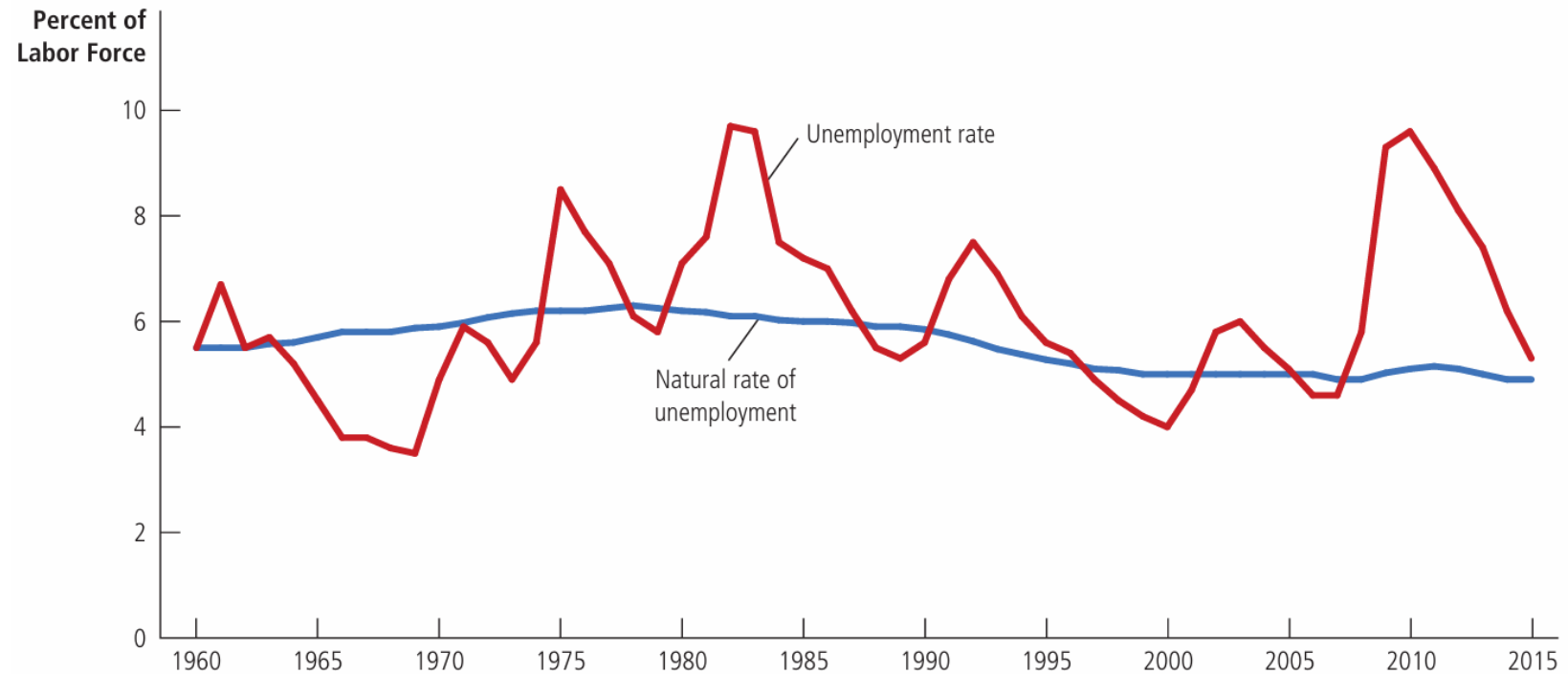
$$\text{LFPR} = \frac{158.3}{252.4} \times 100 = 62.7\%$$

Meaning

- Nearly two-thirds of the adult population participated in the labor market.
- Hence, in January 2016, almost two-thirds of the U.S. adult population were participating in the labor market, and 4.9 percent of those labor-market participants were without work.

IDENTIFYING UNEMPLOYMENT

- Figure 2 shows the unemployment rate in the United States since 1960.
- The figure shows that the economy always has some unemployment and that the amount changes from year to year.



- The normal rate of unemployment around which the unemployment rate fluctuates is called the **natural rate of unemployment**.
- The deviation of unemployment from its natural rate is called **cyclical unemployment**.

IDENTIFYING UNEMPLOYMENT

Does the Unemployment Rate Measure What We Want It To?

- ❖ It is easy to distinguish between a person with a full-time job and a person who is not working at all.
- ❖ But it is much harder to distinguish between a person **who is unemployed** and a person **who is not in the labor force**.
 - Frequent movement in and out of the labor force makes unemployment statistics hard to interpret.
 - Some people **counted as unemployed** may not be seeking work, but instead want to qualify for government benefits.
 - These individuals **may be better classified as out of the labor force** or even employed.
 - Conversely, some **who appear out of the labor force** actually want jobs but have quit searching after repeated failure.
 - These discouraged workers are **not counted in unemployment** figures, even though they lack jobs.

Hence, several other measures might be needed in addition to the official unemployment rate.

IDENTIFYING UNEMPLOYMENT

How Long Are the Unemployed Without Work?

Why Does Duration Matter?

- ❖ The seriousness of unemployment depends not only on **how many people are unemployed**, but also on **how long they remain unemployed**.

Short-Term Unemployment

- Workers may need only a few weeks to find suitable jobs.
- Often reflects normal job-search and labor-market adjustments.
- Generally considered less serious.

Key Question for Economists: Is unemployment typically short-term or long-term?

Long-Term Unemployment

- Workers remain unemployed for many months or longer.
- Can cause:
 - Economic hardship (loss of income)
 - Psychological stress
 - Reduced well-being and confidence
- Generally viewed as a more serious problem.

IDENTIFYING UNEMPLOYMENT

How Long Are the Unemployed Without Work?

A Surprising Finding

What Economists Discovered

- Research on unemployment duration reveals a seemingly contradictory result:
- **Most spells of unemployment are short, but most unemployment observed at any given time is long-term.**

Result

- $52/55 \approx 95\%$ of unemployment spells are short-term.
- Yet most unemployment seen at any point in time comes from the few long-term unemployed workers.

Illustrative Example

- Visit an unemployment office every week for one year.
- Each week, there are **4 unemployed workers**:
 - ✓ **3 workers** remain unemployed for the entire year.
 - ✓ **1 worker** changes every week.
- **Over the Year:** Total unemployed individuals observed = **55**
 - ✓ 52 unemployed for only **1 week**
 - ✓ 3 unemployed for the **entire year**

IDENTIFYING UNEMPLOYMENT

How Long Are the Unemployed Without Work?

Policy Implications

Understanding the Unemployment Problem

- ✓ Although **95% of unemployment spells** are short-term,
- ✓ At any given moment: **75% of observed unemployment** is attributable to the **3 workers unemployed for the entire year**.

Key Insight

- ✓ Most unemployment spells → Short-term
- ✓ Most unemployment observed → Long-term

Implications for Policymakers

- Most unemployed individuals find jobs relatively quickly.
- The bulk of the unemployment problem is concentrated among a **small group of long-term unemployed workers**.
- Effective unemployment policies should pay special attention to workers who remain jobless for extended periods.

Takeaway:

- Most people who become unemployed soon find work, but most of the economy's unemployment burden comes from the relatively few workers who experience long-term unemployment.

IDENTIFYING UNEMPLOYMENT

Why Are There Always Some People Unemployed?

So far, we have learned:

- How unemployment is measured.
- Problems in interpreting unemployment statistics.
- The duration of unemployment spells.

Key Question:

- Why does unemployment exist in the first place?

- ✓ In most markets, prices adjust to balance:

$$\text{Quantity Supplied} = \text{Quantity Demanded}$$

- ✓ Similarly, in an ideal labor market:

$$\text{Labor Supplied} = \text{Labor Demanded}$$

through wage adjustments.

Implication:

- All workers willing to work at the equilibrium wage would be employed.
- Unemployment would disappear.

IDENTIFYING UNEMPLOYMENT

Why Are There Always Some People Unemployed?

The Reality of Labor Markets

Why Doesn't Unemployment Fall to Zero?

- ❑ In reality:
 - Some workers are always unemployed.
 - This remains true even when the economy is performing well.
 - The unemployment rate never reaches zero permanently.
- ❑ Instead, unemployment fluctuates around the natural rate of unemployment.

Natural Rate of Unemployment

Ideal Labor Market



Full Employment

Actual Labor Market



Natural Rate of Unemployment



Persistent Joblessness

Key Insight:

- Actual labor markets differ from the ideal model of full employment.

IDENTIFYING UNEMPLOYMENT

Why Are There Always Some People Unemployed?

Frictional Unemployment

- Workers need time to find jobs that best match their:
 - Skills
 - Experience
 - Preferences
- This matching process creates frictional unemployment.

Frictional Unemployment: Definition

- Unemployment arising from the time required to match workers with suitable jobs.

Characteristics

- ✓ Usually short-term
- ✓ Occurs even in a healthy economy
- ✓ Reflects normal labor-market activity

Worker Leaves Job



Job Search



New Job Found

IDENTIFYING UNEMPLOYMENT

Why Are There Always Some People Unemployed?

Structural Unemployment

- ❑ Sometimes the number of available jobs is insufficient for everyone seeking work.
- ❑ This occurs when:
$$\text{Labor Supply} > \text{Labor Demand}$$
- ❑ The resulting unemployment is called structural unemployment.

Above-Equilibrium Wage



Labor Supply > Labor Demand



Structural Unemployment

Structural Unemployment: Characteristics

- Often long-term
- Caused by wages remaining above equilibrium
- Workers are willing to work but cannot find jobs

Sources of Above-Equilibrium Wages

1. Minimum-Wage Laws
2. Labor Unions
3. Efficiency Wages

Takeaway:

- Long-run unemployment consists of two broad categories: **frictional unemployment** (job search) and **structural unemployment** (insufficient jobs due to above-equilibrium wages).

MINIMUM-WAGE LAWS

From Frictional to Structural Unemployment

- ❑ Previously, we saw that **frictional unemployment** arises because workers need time to find suitable jobs.
- ❑ Now, we examine **structural unemployment**, which occurs when:
 - The number of available jobs is insufficient for the number of workers seeking employment.
- ❑ One important cause of structural unemployment is minimum-wage laws.

Minimum-Wage Laws

- Minimum wages are not the main source of unemployment.
- However, they significantly affect certain groups with high unemployment rates.
- Understanding minimum wages helps explain other forms of structural unemployment as well.

MINIMUM-WAGE LAWS

How Minimum-Wage Laws Create Unemployment

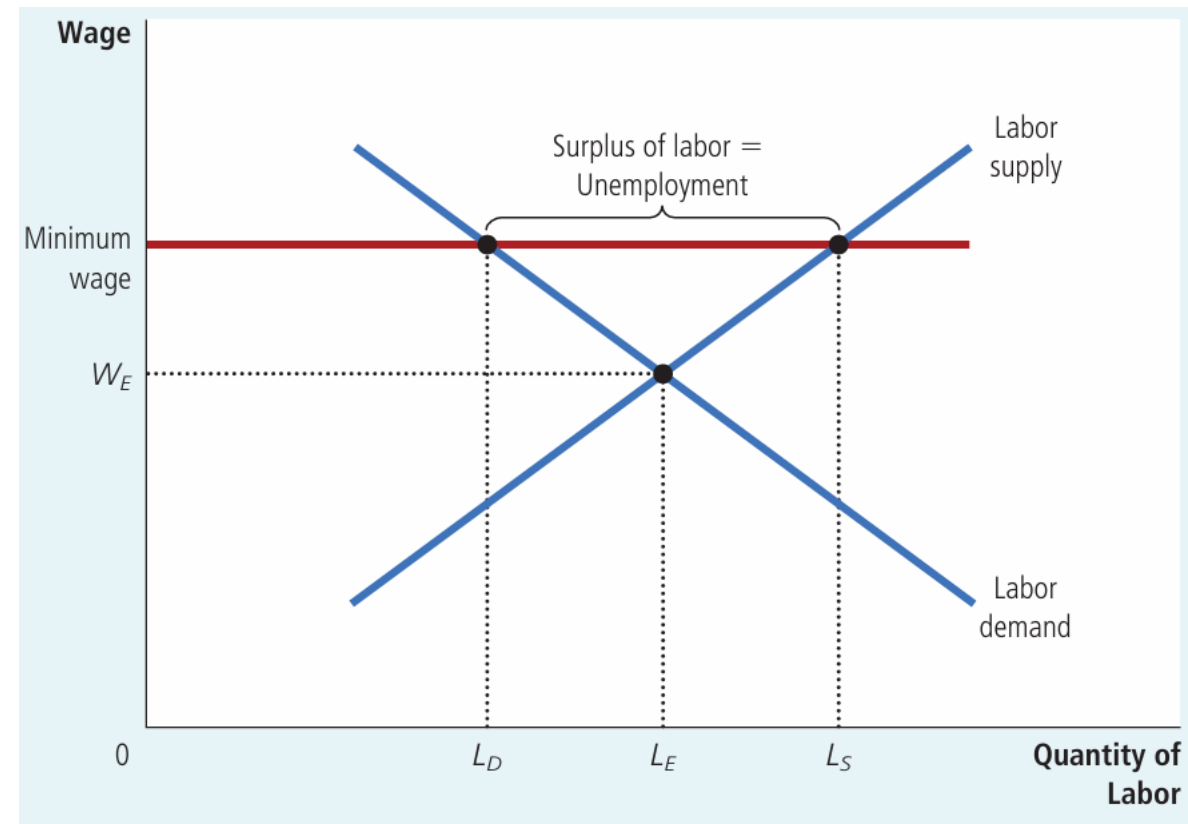
- ❖ When a minimum-wage law sets wages **above the equilibrium wage**:
 $W_{min} > W_E$, the labor market cannot clear.

Effects:

- Quantity of labor supplied increases.
- Quantity of labor demanded decreases.
- A surplus of labor emerges.

Result:

- More workers want jobs than employers are willing to hire, which in turn creates unemployment



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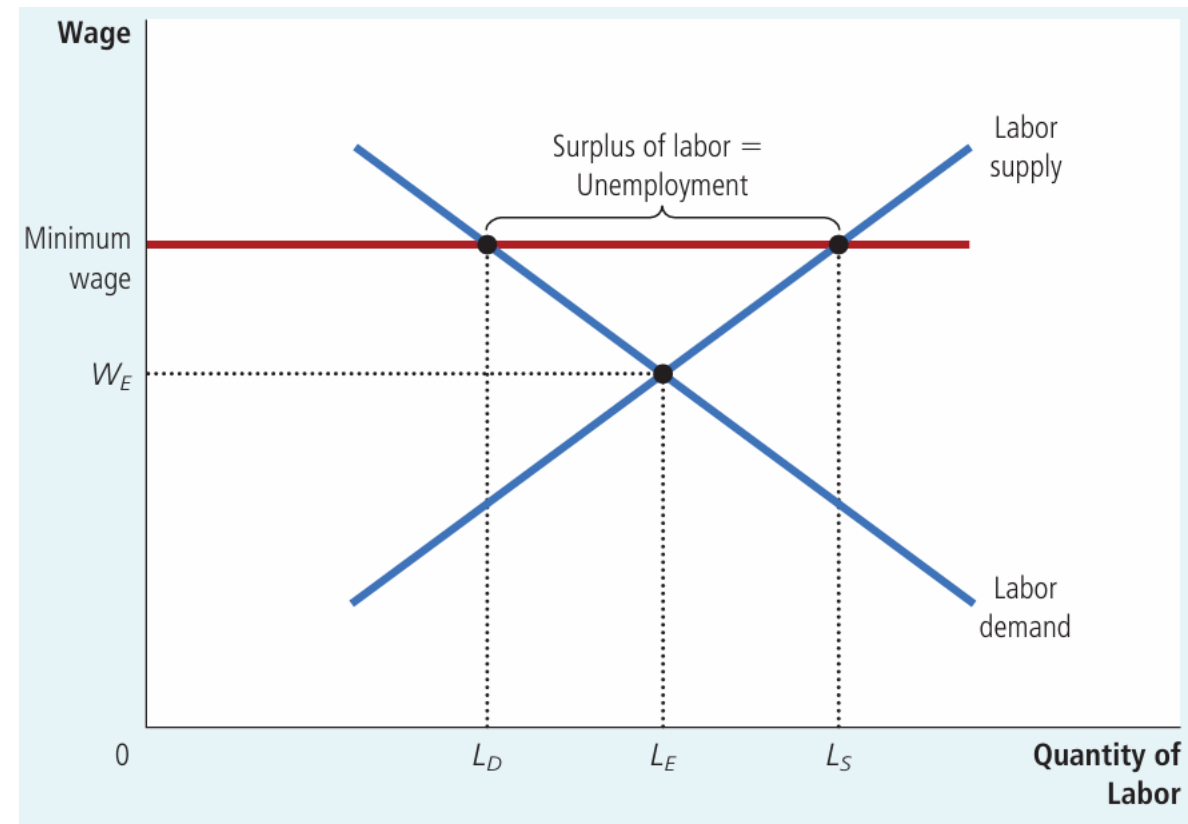
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MINIMUM-WAGE LAWS

Who Is Most Affected?

- Minimum-Wage Laws Do Not Affect Everyone Equally
- Most workers earn wages well above the legal minimum wage.
- Therefore:
 - ✓ Their wages continue to adjust according to supply and demand.
 - ✓ They are largely unaffected by minimum-wage legislation.

Most Affected Groups

- ✓ Least-skilled workers
- ✓ Least-experienced workers
- ✓ Teenagers
- ❖ These workers typically have:
 - Lower productivity
 - Lower equilibrium wages
- ❖ As a result, their market wage is more likely to fall below the legal minimum wage.

Key Point:

- Minimum-wage laws explain unemployment primarily among low-skilled and inexperienced workers.

MINIMUM-WAGE LAWS

Structural vs. Frictional Unemployment

Frictional Unemployment

- Caused by job search.
- Workers are looking for jobs that best match their skills and preferences.
- Labor supply and labor demand are not necessarily out of balance.

Structural Unemployment

- Caused by wages above equilibrium.
- Quantity of labor supplied exceeds quantity demanded.
- Workers are unemployed because they are waiting for jobs to become available.

UNIONS AND COLLECTIVE BARGAINING

What Is a Union?

- A **union** is a worker association that bargains with employers over:
 - ✓ Wages
 - ✓ Benefits
 - ✓ Working conditions

Union Membership

United States

- Only about 11% of workers currently belong to unions.
- During the 1940s and 1950s, about **one-third of the labor force** was unionized.

Key Point

- ❖ In unionized labor, wages are often determined through collective bargaining rather than purely by supply and demand.

Europe

- Unions remain highly.
- In **Belgium, Norway, and Sweden**, more than half of workers belong to unions.
- In **France and Germany**, many workers have wages determined through collective bargaining, even if they are not union members.

UNIONS AND COLLECTIVE BARGAINING

The Economics of Unions

Unions as a Labor Cartel

- ❑ A union can be viewed as a **cartel of workers**.

Individual Bargaining vs Collective Bargaining

- Without Union: Worker ↔ Employer
- With Union: Workers (Union) ↔ Employer

Collective Bargaining

- The process through which unions and firms negotiate:
 - Wages
 - Benefits
 - Working conditions

Strike Power

- If negotiations fail, workers may organize a **strike** (withdrawal of labor).
- Because strikes reduce:
 - ✓ Production
 - ✓ Sales
 - ✓ Profits, firms often agree to higher wages.

Evidence:

- Union workers typically earn about **10–20% more** than similar nonunion workers.

UNIONS AND COLLECTIVE BARGAINING

How Unions Affect the Labor Market

- When unions negotiate wages above the equilibrium level: $W_{Union} > W_E$

Consequences

Higher Union Wage



Labor Supply Increases



Labor Demand Falls



Unemployment

Winners and Losers

Insiders

- ✓ Workers who keep union jobs.
- ✓ Receive higher wages and benefits.

Outsiders

- ✓ Workers unable to obtain union jobs.
- ✓ May remain unemployed or seek work elsewhere.

UNIONS AND COLLECTIVE BARGAINING

Spillover Effects of Union Wages

- ❑ When outsiders cannot obtain union jobs:
 1. Some remain unemployed.
 2. Others move to nonunion sectors.
- ❑ As labor supply increases in nonunion industries:

Labor Supply ↑
→ Nonunion wages may fall.
- ❑ **Result:**
 - Union members gain from collective bargaining, while some nonunion workers bear part of the cost.

UNIONS AND COLLECTIVE BARGAINING

Are Unions Good or Bad for the Economy?

Critics' View: Unions Harm Economic Efficiency

- ❑ Critics argue that unions are essentially a **labor cartel**.
- ❑ When unions raise wages above the competitive market level:
 - ✓ Labor demand decreases.
 - ✓ Some workers become unemployed.
 - ✓ Wages in nonunion sectors may fall.

Economic Consequences

Inefficient

- Employment in unionized firms falls below the competitive level.
- Labor resources are not allocated efficiently.

Inequitable

- ✓ Union workers benefit from higher wages.
- ✓ Other workers may lose jobs or face lower wages.

UNIONS AND COLLECTIVE BARGAINING

Are Unions Good or Bad for the Economy?

Advocates' View: Unions Protect Workers

- ❑ Advocates argue that unions help balance the market power of employers.
- ❑ **Example: Company Town**
 - In a "company town":
 - One firm dominates local employment.
 - Workers have few alternative job opportunities.
 - Employers may offer lower wages and poorer working conditions.
 - In such cases, unions can:
 - ✓ Increase workers' bargaining power.
 - ✓ Protect employees from employer dominance.
 - ✓ Improve wages and working conditions.

Additional Benefits

- ❖ Unions help firms respond to workers' concerns regarding:
 - Working hours
 - Overtime
 - Vacations
 - Sick leave
 - Health benefits
 - Promotion opportunities
 - Job security

Key Takeaway

- ❖ There is no consensus among economists about whether unions are good or bad for the economy. Their effects may be beneficial in some circumstances and harmful in others.